

Week 5

Physics
stack
rebuild

New physics stack and the debugging path

4H-SiC LGAD TCAD | Week 5 update

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Honest status:
no results yet, but the deck is rebuilt

Where we are

Week 4 ending point

- ▶ Multi-zone JTE geometry built
- ▶ Toy SEU transient at 1000 V
- ▶ Staircase I-V, no clean avalanche
- ▶ `conmob` on SiC (wrong)
- ▶ No precision flag, no `ni.min`
- ▶ No traps, no acceptor removal

Week 5 status

- ▶ No converged results yet
- ▶ Full physics stack rebuilt
- ▶ Every parameter traced to a paper
- ▶ Every crash diagnosed and documented
- ▶ Two-step initialization designed
- ▶ Impact deferral strategy identified

What changed this week

1. **Two-step initialization** — lightweight models for the initial guess, heavy physics for the sweep
2. **Two-level incomplete ionization** — `inc.two_level` for Al on cubic and hexagonal sites
3. **Fermi-Dirac with `ni.fermi`** — self-consistent n_i from the Fermi level, not the empirical value
4. **Lindefelt BGN** — `bgn.lindefelt` calibrated for 4H-SiC, replacing the silicon-default `bgn`
5. **Hatakeyama + `gradqfl` + `e.side`** — anisotropic impact ionization with quasi-Fermi driving field
6. **Trap-Assisted Tunneling for $Z_{1/2}$** — `tat.trap` on the defect, `trap.tunnel` in the models
7. **Interface: no anneal** — `qf=1e11`, `s.n=s.p=1e4`, matching onsemi's as-oxidized SiO₂/SiC
8. **256-bit precision + tightened tolerances** — the SiC gold standard

Each change is documented in the *Onward* mentor guide (Part IV, Stage 0) and the *Peak Decks V2* handbook (Part III, Chapters 9–15).

Two-step initialization

Step 1: lightweight (for the initial guess)

```
# Lightweight physics
models material='4H-SiC' \
    consrh auger analytic fldmob \
    bgn.lindefelt

method newton norm.scaling.local \
    itlimit=100 dvmax=10 weak=1 \
    climit=1e-4

solve init
```

Step 2: heavy physics (for the real sweep)

```
models material='4H-SiC' \
    fermidirac ni.fermi \
    incomplete inc.two_level \
    bgn.lindefelt consrh auger \
    analytic fldmob trap.tunnel \
```

Why this works

Newton-Raphson finds the root of the PDE residual. The lightweight step gets the solver 95% there. The heavy step walks the remaining 5% and settles into the *same unique solution* it would have found otherwise.

What I was doing wrong

Turning on incomplete inc.two_level *before* solve init. At 0V, deep Al dopants cause infinite oscillations in the Fermi equilibrium solver. One run took 30 minutes at init and never converged.

Two-level incomplete ionization

The physics

$$N_A^- = \frac{N_A}{1 + g_A \exp\left(\frac{E_A - E_F}{k_B T}\right)}$$

- ▶ Al acceptor: $E_A \approx 0.26$ eV above E_V
- ▶ N donor: $E_D \approx 0.06$ eV below E_C
- ▶ At 300 K, only ~ 10 – 20% of Al is ionised
- ▶ Two-level: cubic (k) and hexagonal (h) sites
- ▶ $E_A(k) = E_A(h) = 0.32$ eV; $E_D(k) = 0.09$,
 $E_D(h) = 0.05$ eV

The deck syntax

```
models material='4H-SiC' \  
  fermidirac ni.fermi \  
  incomplete inc.two_level \  
  bound.trap \  
  acc.inc.dynamic \  
  don.inc.dynamic
```

Why bound.trap

Required with incomplete. Couples bound trap state to SRH recombination.

Victory Device manual Table 3-16; Castaldini *et al.*, JAP 98, 053706 (2005).

Fermi-Dirac with `ni.fermi`

Why Boltzmann fails in 4H-SiC

$$f(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{k_B T}\right)}$$

- ▶ Boltzmann: $f \approx \exp(-(E - E_F)/kT)$
- ▶ Valid only when E_F is $\gg kT$ from band edge
- ▶ In 4H-SiC at 243 K: $n_i \sim 10^{-20} \text{ cm}^{-3}$
- ▶ Fermi level pinned near E_C or E_V in doped regions
- ▶ Boltzmann gives wrong n, p by orders of magnitude

The two-stage flag

```
models material='4H-SiC' \  
    fermidirac ni.fermi ...
```

fermidirac

Switches from Boltzmann to Fermi-Dirac. Mandatory for wide-bandgap.

ni.fermi

Computes n_i self-consistently from the Fermi level. At 243 K the empirical $n_i \approx 0$, causing divide-by-zero. `ni.fermi` fixes this.

Bandgap narrowing: Lindefelt model

What BGN does

$$E_g^{\text{eff}} = E_g - \Delta E_g(N)$$

- ▶ In the p^{++} anode (10^{19} cm^{-3}):
 $\Delta E_g \sim 80 \text{ meV}$
- ▶ In the n^{++} substrate (10^{20} cm^{-3}):
 $\Delta E_g \sim 120 \text{ meV}$
- ▶ In the drift region (1.5×10^{14}): negligible
- ▶ Affects contact injection and the effective n_i

Why Lindefelt, not Slotboom

- ▶ bgn (default): Slotboom-Cornell, calibrated for Si
- ▶ bgn.lindefelt: Lindefelt, calibrated for 4H-SiC and 6H-SiC
- ▶ Treats n -type and p -type BGN separately
- ▶ Has 4H-SiC default coefficients built in

```
models material='4H-SiC' \  
... bgn.lindefelt ...
```

Lindefelt, JAP 76, 3768 (1994). Also used in Silvaco SiC_Power_ex10/ex11.

Impact ionization: Hatakeyama + gradqfl + e.side

The model: Hatakeyama anisotropic

$$\alpha(E) = A \exp\left(-\frac{B}{E}\right)$$

- ▶ Separate A , B for $\langle 0001 \rangle$ and $\langle 11\bar{2}0 \rangle$
- ▶ $A_{e,0001} = 1.76 \times 10^8 \text{ cm}^{-1}$
- ▶ $B_{e,0001} = 3.30 \times 10^7 \text{ V/cm}$
- ▶ Built in as SIC4H0001
- ▶ In 4H-SiC: hole $\beta >$ electron α (opposite of Si)

The two flags I was missing

```
impact material='4H-SiC' \  
  aniso sic4h0001 \  
  e.side gradqfl \  
  e.min=1e5
```

gradqfl

Drives impact ionization by $|\nabla \phi_n|$, not $|E|$. Essential when current flows — without it, LGAD gain is wrong. From SiC_Power_ex10/ex11.

e.side

Evaluates E at element sides, not nodes. More accurate peak field, prevents Newton crashes. Does *not* mean “electron side”.

Trap-Assisted Tunneling for $Z_{1/2}$

The $Z_{1/2}$ defect

- ▶ Carbon vacancy (V_C), negative- U
- ▶ Acceptor at $E_C - 0.63$ eV
- ▶ Dominant deep level in irradiated 4H-SiC
- ▶ $\sigma_n = 10^{-14} \text{ cm}^2$, $\sigma_p = 10^{-14} \text{ cm}^2$
- ▶ Degeneracy factor: 12

The two-part handshake

```
# 1. The defect (with TAT flag)
trap material='4H-SiC' acceptor \
    density=1e13 \
    energy.level=0.63 \
    degeneracy.factor=12 \
    sign=1e-14 sigp=1e-14 \
```

The 9-hour crash

trap.tunnel forces Newton to calculate quantum-tunneling integrals at every mesh node, every iteration. At 4 V the field is too low for tunneling, so the 17 minutes per step was spent calculating zero. At 1062 V the JTE corner field gradient broke the Jacobian.

The decoupled plan

Deck A (Avalanche): no traps, fast sweep to V_{BD} . **Deck B (Defects):** traps + TAT, sweep only to ~ 200 V. **Python:** stitch the avalanche curve and the leakage floor together.

Castaldini *et al.*, JAP 98, 053706 (2005). Hurkx TAT model.

Interface: why *no* anneal for onsemi

The onsemi interface

```
interface s.i \  
  qf=1e11 \  
  s.n=1e4 s.p=1e4
```

- ▶ $qf=1e11$: fixed oxide charge (cm^{-2})
- ▶ $s.n=s.p=1e4$: surface recombination (cm/s)
- ▶ As-oxidized — **no** post-oxidation anneal

Why no anneal in SiC (unlike Si)

- ▶ In Si: anneal *reduces* interface traps
- ▶ In 4H-SiC: anneal *increases* NITs
- ▶ Carbon clusters form during oxidation
- ▶ Annealing drives more C into clusters
- ▶ onsemi keeps as-oxidized interface

My previous mistake

I tuned SRV as a toy number ($1000 \rightarrow 10 \rightarrow 10000$) to force 10^{-11} A leakage. In pure SiC ($n_i \sim 10^{-9}$), no SRV produces thermal leakage. The 10^{-11} A floor is pad-perimeter + SMU noise, not bulk SRH.

Precision: 256-bit with tightened tolerances

Why 64-bit fails for SiC

- ▶ $n_i \sim 10^{-8} \text{ cm}^{-3}$ at 300 K
- ▶ $N_D \sim 10^{15} \text{ cm}^{-3}$ in the drift
- ▶ Ratio $N_D/n_i \sim 10^{23}$
- ▶ 64-bit float: ~ 16 digits
- ▶ Newton loses precision near BV

Why 256-bit

- ▶ Gold standard for 4H-SiC
- ▶ Example decks: up to 256-bit (ex02)
- ▶ First run failed: tolerances not tightened

Silvaco SiC_Power_ex02: precision sweep 80/128/160/256-bit.

The deck syntax

```
go victorydevice simflags="-256 -P 4"

# Tolerances MUST tighten
method newton norm.scaling.local \
    itlimit=100 dvmax=10 weak=1 \
    climit=1e-4 \
    px.tol=1e-22 cx.tol=1e-22 \
    pr.tol=1e-43 cr.tol=1e-35
```

The lesson

256-bit *without* tightened tolerances = wasted precision. At
256-bit: px.tol=1e-22, pr.tol=1e-43.

Newton method: dvmax and the two-card pattern

Card 1: general (pre-breakdown)

```
method newton norm.scaling.local \  
  itlimit=100 dvmax=10 weak=1 \  
  climit=1e-4 \  
  px.tol=1e-12 cx.tol=1e-12 \  
  pr.tol=1e-22 cr.tol=1e-22
```

- ▶ dvmax=10: large steps, system not stiff
- ▶ itlimit=100: generous budget

Card 2: tightened (near breakdown)

```
method newton norm.scaling.local \  
  itlimit=50 dvmax=2.5 weak=1 \  
  climit=1e-4 \  
  px.tol=1e-12 cx.tol=1e-12 \  
  pr.tol=1e-22 cr.tol=1e-22
```

- ▶ dvmax=2.5: prevent overshoot near BV
- ▶ itlimit=50: fail fast

The Newton bounce

At 1056 V, solver found the solution at iteration 17 but `pr.tol=1e-20` was too greedy. It hit a floating-point wall and exploded. Fix: relax to `pr.tol=1e-14`.

Parameters I: material, mobility, recombination

1 0.85

Material constants

Parameter	Value	Source
E_g (300)	3.25 eV	Harris (1995); ex05
Varshni α	9.06×10^{-4} eV/K	ex05
Varshni β	2.03×10^3 K	ex05
Affinity	3.13 eV	ex06
N_C (300)	1.66×10^{19} cm $^{-3}$	Harris (1995)
N_V (300)	3.3×10^{19} cm $^{-3}$	Harris (1995)
ϵ_r	9.7	Patrick-Choyke (1970)
ni.min	2 cm $^{-3}$	ex10 (numerical floor)

Mobility

(Caughey-Thomas)

Parameter	Value	Source
$\mu_{n,0}$	947 cm 2 /Vs	Caughey-Thomas; ex06
$\mu_{p,0}$	124 cm 2 /Vs	ex06 (mu2p.caug)
$N_{ref,n}$	1.94×10^{17} cm $^{-3}$	Caughey-Thomas
$N_{ref,p}$	1.76×10^{19} cm $^{-3}$	ex06
δ_n (T exp.)	0.61	Caughey-Thomas
δ_p (T exp.)	0.34	ex06
$v_{sat,n}$	2.2×10^7 cm/s	Hatakeyama (2011)
$v_{sat,p}$	1.0×10^7 cm/s	VERIFY: 0.8–1.2e7

Recombination

Parameter	Value	Source
τ_{n0} (SRH)	1×10^{-6} s	Device-grade baseline
τ_{p0} (SRH)	1×10^{-6} s	Device-grade baseline
C_n (Auger)	5×10^{-29} cm 6 /s	Galeckas (1997)
C_p (Auger)	9.9×10^{-32} cm 6 /s	Ščajev
BGN model	Lindefelt	Lindefelt, JAP (1994)

Incomplete

ionization

Parameter	Value	Source
Model	Two-level	Victory Device default
Al E_A (cubic)	0.32 eV	Manual Table 3-16
Al E_A (hex)	0.32 eV	Manual Table 3-16
N E_D (cubic)	0.09 eV	Manual Table 3-18
N E_D (hex)	0.05 eV	Manual Table 3-18
Al degeneracy g_A	4	Standard
N degeneracy g_D	2	Standard

Parameters II: impact, traps, interface, numerical

1 0.85

Impact ionization (Hatakeyama)

Parameter	Value	Source
Model	aniso sic4h0001	Hatakeyama (2011)
$A_{e,0001}$	$1.76 \times 10^8 \text{ cm}^{-1}$	Built-in
$B_{e,0001}$	$3.30 \times 10^7 \text{ V/cm}$	Built-in
$A_{h,0001}$	$3.41 \times 10^8 \text{ cm}^{-1}$	Built-in
$B_{h,0001}$	$2.50 \times 10^7 \text{ V/cm}$	Built-in
$A_{e,11-20}$	$2.10 \times 10^7 \text{ cm}^{-1}$	Built-in
$B_{e,11-20}$	$1.70 \times 10^7 \text{ V/cm}$	Built-in
e.side	ON	Element-side E eval
gradqfl	ON	Quasi-Fermi driving field
e.min	10^5 V/cm	Low-field cutoff

Trap ($Z_{1/2}$)

Parameter	Value	Source
Type	Acceptor	Castaldini (2005)
Energy	$E_C - 0.63 \text{ eV}$	Castaldini (2005)
Density	10^{13} cm^{-3}	To be fluence-scaled
σ_n	10^{-14} cm^2	Castaldini (2005)
σ_p	10^{-14} cm^2	Castaldini (2005)
Degeneracy	12	Negative- U V_C
tat.trap	ON	Hurkx TAT

Interface (SiO_2/SiC)

Parameter	Value	Source
Q_f (fixed oxide)	10^{11} cm^{-2}	onsemi as-oxidized
S_n (surf. recomb.)	10^4 cm/s	onsemi as-oxidized
S_p (surf. recomb.)	10^4 cm/s	onsemi as-oxidized
Anneal	NO	onsemi process

Numerical method

Parameter	Value	Source
Precision	256-bit	ex02 (gold standard)
Solver	Newton	—
norm.scaling.local	ON	ex10 (mandatory SiC)
climit	10^{-4}	ex01
dvmax (general)	10	ex10
dvmax (BV sweep)	2.5	ex10
px.tol	10^{-22}	ex02/ex11
pr.tol	10^{-43}	ex02
weak	1	ex01

Full parameter reference: *Onward guide* Appendix I; *Peak Decks V2* Appendix F.

The debugging journey

Crash	Cause	Lesson
30 min at solve init	<code>incomplete inc.two_level</code> ON before init	Two-step initialization: lightweight models first, heavy physics after
9 hours, crash at 1062 V	<code>trap.tunnel</code> quantum integrals at every node	Decouple: avalanche deck (no traps) + defect deck (traps, low V)
Current at 10^{-33} A	Pure SiC has $n_i \sim 10^{-9}$; no thermal leakage	10^{-11} A floor is pad-perimeter + SMU noise, not TCAD
Newton bounce at 1056 V	<code>pr.tol=1e-20</code> too greedy	Relax to <code>pr.tol=1e-14</code> ; accept the solution at iteration 17
Cut back $7\times$ at 225 V	Gain-layer depletion punch-through snap	Impact deferral: sweep past 232 V with impact OFF, on at 400 V
17.5 min/step at 4 V	<code>trap.tunnel</code> calculating zero tunneling at low field	TAT only in the defect deck, not the avalanche deck
Condition number explosion	<code>con.resist=1e-4</code> + pure SiC = divide by zero	Do not combine contact resistance with SiC at 0 V
<code>autonr</code> warning	Legacy Atlas command, not Victory Device	Use <code>maxtraps=15</code> instead

The pattern Every crash was a numerical issue, not a physics issue. The physics stack is now correct. The remaining work is

Next steps

- ▶ Run the decoupled avalanche deck (no traps, two-step init, impact deferral) to get a clean V_{BD}
- ▶ Run the defect deck with $Z_{1/2}$ + trap-assisted tunneling to get the leakage floor, then stitch with the avalanche curve in Python
- ▶ Calibrate the IV/CV against onsemi data and begin the impact-ionization benchmark (Hatakeyama vs. Konstantinov vs. Loh)